

NAND to Tetris-I

Detailed Assessment & Submission Template

Single-PDF submission checklist and assessment format

1. Student / Submission Details

Student Name	
Roll / Registration No.	
Programme / Semester	
Course / Subject	NAND to Tetris-I
Assessment / Task	Memory Hierarchy / Cache Design Assessment
Date of Submission	

2. Assessment Objective

Demonstrate a systematic understanding of memory hierarchy by implementing/documenting the required HDL logic, verifying operation with timing diagrams, decomposing addresses, comparing latency, documenting state transitions, and relating the design to real L1/L2 cache systems.

3. Mandatory Deliverables

- ☐ HDL code listings
- ☐ Timing diagrams / simulation evidence
- ☐ Address decomposition calculations
- ☐ Latency comparison chart
- ☐ State-transition table
- ☐ Brief reflection connecting the model to real L1/L2 cache hierarchy
- ☐ A single, clearly organized PDF containing all work
- ☐ Final PDF checked for readability, completeness, page order, and correct file format

4. Recommended Assessment Rubric — 100 Marks

Criterion	Marks	What is assessed
A. Problem understanding & design assumptions	10	Clear interpretation, assumptions, architecture/block-level explanation
B. HDL implementation	25	Correct HDL, modular structure, signals, logic, comments, synthesis/simulation readiness
C. Functional verification	15	Test cases, expected vs actual behaviour, timing/simulation evidence
D. Address decomposition	15	Correct tag/index/offset or equivalent decomposition, calculations and examples
E. Latency comparison	10	Meaningful comparison, units,

		hit/miss cases, chart and explanation
F. State-transition table	10	Complete states, inputs/events, next states and outputs/actions
G. L1/L2 cache reflection	10	Technically accurate connection to real cache hierarchy
H. Presentation & PDF quality	5	Organization, labels, captions, readability and single-PDF compliance
TOTAL	100	All required components completed

5. Section A — Problem Definition & Design Assumptions

Write ½–1 page covering:

- What memory hierarchy / cache behaviour is being modelled?
- What are the memory size, block/line size, address width and data width?
- What assumptions are made about read/write behaviour?
- What happens on a hit?
- What happens on a miss?
- What information is stored in each cache entry?
- What simplifications are made compared with a real processor cache?

Student response:

6. Section B — HDL Code Listing

Include the complete HDL code used for the assessment. Do not paste unexplained fragments.

- ☐ Module/component name
- ☐ Inputs and outputs with widths
- ☐ Internal registers/wires
- ☐ Combinational logic
- ☐ Sequential/state logic
- ☐ Reset/initialization behaviour
- ☐ Read/write control
- ☐ Hit/miss detection, if applicable

☐ Comments explaining important design decisions

☐ Testbench code, if used

HDL listing / code pages: _____

Verification status: ☐ Pass ☐ Needs correction

Instructor comments: _____

7. Section C — Timing Diagrams & Verification

For every important operation, provide a labelled timing diagram or simulation waveform.

Test case	What must be shown	Status
Reset / initialization	Show reset, initial state and release	☐ Pass ☐ Revise
Read hit	Address → lookup → hit → data output	☐ Pass ☐ Revise
Read miss	Address → miss detection → lower-memory access → refill → output	☐ Pass ☐ Revise
Write operation	Show control signals and resulting memory/cache update	☐ Pass ☐ Revise
Boundary / corner case	Use an address near an index/block boundary	☐ Pass ☐ Revise

For each waveform: label clock, reset, address, data, control signals, hit/miss and state where relevant.

8. Section D — Address Decomposition

Show the complete calculation. Do not provide only the final numbers.

- Total address width = _____ bits
- Memory capacity = _____
- Cache capacity = _____
- Block/line size = _____ bytes
- Number of cache lines/sets = _____
- Offset bits = _____
- Index/set bits = _____
- Tag bits = _____
- Check: tag + index + offset = total address bits

Worked example address: _____

Binary address: _____

Tag: _____ Index/Set: _____ Offset: _____

Verification of decomposition: _____

9. Section E — Latency Comparison

Prepare a chart comparing the access time for the relevant hierarchy levels.

Access type	Latency (cycles)	Latency (ns)	Observation
Cache hit			
Cache miss → next level			
L1 access			
L2 access			
Main memory access			

Required chart: ☐ Bar chart ☐ Table + chart ☐ Other: _____

Explain why a cache hit is faster than accessing a lower memory level.

10. Section F — State-Transition Table

Use one row for every meaningful state/event combination.

Current State	Input/Event	Condition	Next State	Output/Action

Suggested states (modify according to your design): IDLE, LOOKUP, HIT, MISS, REFILL, UPDATE, COMPLETE.

11. Section G — Real L1/L2 Cache Reflection

Write approximately 300–500 words. Address all points below:

- How does your model represent locality of reference?
- How is L1 different from L2 in purpose and typical latency?
- Why are higher cache levels generally larger but slower?
- What does a cache hit mean in practical processor operation?
- What happens when data is not found in L1 but may be present in L2?
- What important real-world features are simplified or absent in your model (e.g., associativity, replacement policy, coherence, write policy, multi-core effects)?
- What did you learn from implementing the model rather than only studying the theory?

Reflection:

12. Section H — Final Verification Checklist

- ☐ All HDL modules compile without errors
- ☐ Testbench/simulation runs successfully
- ☐ Timing diagrams are readable and correctly labelled
- ☐ At least one hit and one miss are demonstrated
- ☐ Address decomposition is mathematically verified
- ☐ Latency values have units and a clear basis
- ☐ Latency chart is included
- ☐ State-transition table is complete
- ☐ Reflection is technically accurate and written in the student's own words
- ☐ Figures/tables have numbers and captions
- ☐ All HDL code is included in the PDF
- ☐ No important page is cut off or unreadable
- ☐ PDF contains all required work in one file
- ☐ Final PDF opens correctly and is within the 25 MB limit
- ☐ File is ready for upload to ZOSIMA

13. Suggested Final PDF Order

1. Cover page
2. Problem statement / objective
3. Design assumptions and architecture
4. HDL code listings
5. Testbench and verification results
6. Timing diagrams
7. Address decomposition and worked examples
8. Latency comparison table and chart
9. State-transition table
10. Reflection: model vs real L1/L2 cache
11. Conclusion / key learning
12. Final verification checklist

14. Final Submission Information

Final PDF filename	
PDF size	_____ MB (must be ≤ 25 MB)
Number of pages	_____
Upload platform	ZOSIMA
Submission status	☐ Ready ☐ Needs revision

Important: The uploaded problem statement requires one PDF containing HDL code listings, timing diagrams, address decomposition workings, a latency comparison chart, a state-transition table, and a brief reflection on how the memory hierarchy relates to real L1/L2 cache.